

Conference on Algebraic Representation Theory 2026

CART 2026

December 4 (Fri.) – 6 (Sun.), 2026

Okinawa Institute of Science and Technology (OIST)
Seminar Room L4E48, Okinawa, Japan

Titles and Abstracts

Speakers are listed in alphabetical order.

Katsuyuki Bando (Keio University)

Title: Derived geometric Satake equivalence in mixed characteristic

Abstract: The geometric Satake equivalence is the categorification of the Satake isomorphism. Its derived version, “the derived geometric Satake equivalence,” is proved by Bezrukavnikov–Finkelberg in the case of equal characteristic. In this talk, we will explain a new affine Grassmannian which connects an equal characteristic affine Grassmannian and Zhu’s Witt-vector affine Grassmannian. As a result, we deduce the derived Satake equivalence in mixed characteristic. By the same argument, we also obtain the mixed characteristic version of Bezrukavnikov’s equivalence between two categorifications of an affine Hecke algebra. Moreover, we explain the derived geometric Satake equivalence on the Beilinson–Drinfeld Grassmannian with one leg in mixed characteristic.

Dong Jun Choi (Seoul National University)

Title: Poisson Structures of Deformed and Classical W -algebras

Abstract: Deformed W -algebras have close connections with quantum affine algebras and q -characters. In type A , their Poisson versions admit a description in terms of q -difference operators, paralleling the description of classical W -algebras by differential operators.

In this talk, we present an algebraic formulation of q -difference Drinfeld–Sokolov reduction and explain how it produces the deformed W -algebra of type A . Using determinant formulas and the Miura transformation, we construct explicit linear combinations of the deformed W -algebra generators whose formal limits recover the standard generators of the classical W -algebra. We also show that the Poisson brackets recover the classical Poisson structure under this limit.

This talk is based on joint work with Uhi Rinn Suh.

Jiepeng Fang (The University of Hong Kong)

Title: Thickening realization and positivity properties of canonical bases

Abstract: For any symmetric Cartan datum, Lusztig conjectured that the canonical basis $\dot{\mathbf{B}}$ of the modified quantum group $\dot{\mathbf{U}}$ has positivity properties with respect to the multiplication and comultiplication. We introduce an algebraic method called the thickening realization to relate $\dot{\mathbf{B}}$ to the canonical basis of the negative part $\tilde{\mathbf{U}}^-$ of a larger quantum group. More precisely, it identifies the structure constants in $\dot{\mathbf{U}}$ with relevant structure constants in $\tilde{\mathbf{U}}^-$. We prove that $\dot{\mathbf{B}}$ has the positivity properties with respect to the comultiplication and bilinear form of $\dot{\mathbf{U}}$, as well as the multiplication whenever one factor is spherical parabolic. In particular, Lusztig’s positivity conjecture holds for any simply-laced finite type. This is a joint work with Xuhua He.

You-Hung Hsu (National Cheng Kung University)

Title: TBA

Abstract: TBA

Woo-Seok Jung (University of Seoul)

Title: A Comparison of Cluster Algebra Structures Arising from i -Boxes and Demazure Weaves

Abstract: We compare two cluster algebra structures associated with a positive braid of finite ADE type. The first arises from a localized bosonic extension, whose initial seeds are constructed from admissible chains of i -boxes and are closely related to monoidal categorification. The second arises from the coordinate ring of a braid variety, with initial seeds constructed from Demazure weaves.

For each admissible chain of i -boxes, we explicitly construct a corresponding Demazure weave and show that the resulting seeds are identified under an algebra isomorphism between the localized bosonic extension and the coordinate ring of the braid variety. Under this isomorphism, the PBW vectors are identified with the natural coordinates on the braid variety.

This is joint work with JiSun Huh, Myungho Kim, and Euiyong Park.

Chun-Ju Lai (Academia Sinica)

Title: Foundations and Applications of Quantum Wreath Products

Abstract: Quantum wreath products deform classical wreath product groups by allowing relation coefficients to take values in a tensor product algebra rather than the ground field. This framework unifies the study of structure and representation theory of algebras admitting a Bernstein–Lusztig presentation, including Vignéras’ pro- p Hecke algebras for p -adic groups (and their mteaplectic analogs), Kleshchev–Muth’s affine zigzag algebras, and Hu’s “Hecke algebras” for the wreath products between symmetric groups. We will discuss recent progress of our theory in seemingly unrelated directions: the Ginzburg–Guay–Opdam–Rouquier conjecture on highest-weight covers, the realization of imaginary strata for affine KLR algebras in positive characteristic, the breakdown of the local Shimura correspondence at the pro- p level, and the categorification perspectives of towers of quantum wreath products. The talk is based on joint work with Buciumas (Pohang), Cailan Li (Academia Sinica), Minets (Bonn), Nakano (Georgia), and Ziqing Xiang (Shenzhen).

Donggun Lee (Korea Institute for Advanced Study)

Title: TBA

Abstract: TBA

Kang Lu (Southern University of Science and Technology)

Title: TBA

Abstract: TBA

Haruto Murata (The University of Tokyo)

Title: Quantum imaginary Schur–Weyl duality

Abstract: Khovanov–Lauda–Rouquier algebras of affine type A depend on a parameter t and, at $t = 1$, are isomorphic to affine Hecke algebras of type A . The representation theory of Khovanov–Lauda–Rouquier algebras is reduced to the study of quotient algebras defined for each positive root of affine type A via stratification, among which those for imaginary roots are substantially more complicated than those for real roots. When $t = 1$, the imaginary strata are related to Schur algebras through the imaginary Schur–Weyl duality established by Kleshchev and Muth. In this talk, we

generalize this duality to an arbitrary parameter t and establish a connection with t -Schur algebras. As an application, we explain how irreducible characters are related to the canonical basis.

Yung-Ning Peng (National Cheng Kung University)

Title: Shifted twisted Yangians and finite W -algebras

Abstract: It is well known that the finite W -algebra associated to an arbitrary nilpotent element in a general linear Lie algebra is isomorphic to the corresponding truncated shifted Yangian due to Brundan–Kleshchev. Such an isomorphism is known to exist for BCD types of rectangular nilpotent elements, in which the role of the Yangians should be replaced by the corresponding *twisted* Yangians.

In this talk, we will introduce the notion of parabolic presentations for twisted Yangians, which interpolate between the R-matrix and Drinfeld presentations. This allows one to define the *shifted* twisted Yangians and their quotient algebras known as truncated shifted twisted Yangians. We further prove that our truncated shifted twisted Yangian is isomorphic to the finite W -algebra associated with every even nilpotent element in type B and C, as well as some special cases of type D. We conjecture the existence of a certain Pfaffian-like element in finite W -algebras of type D, whose existence will imply that the isomorphism mentioned above exists for type D as well. This talk is based on joint work with Kang Lu, Lukas Tappeiner, Lewis Topley, and Weiqiang Wang.

Heizo Sakamoto (The University of Tokyo)

Title: TBA

Abstract: TBA

Pavel Turek (Okinawa Institute of Science and Technology)

Title: New decomposition columns in every block

Abstract: A central open problem in the modular representation theory of symmetric groups (and Hecke algebras) is the computation of decomposition numbers, which describe how characteristic-zero irreducible modules decompose in prime characteristic. These numbers are organised in a block matrix, called the decomposition matrix, whose structure remains a mystery. In this talk, we present new whole columns of this matrix labelled by a novel family of partitions called d -balanced partitions; this seems to be the first description of whole columns in each block to date! Moreover, the column description is exceptionally elementary. The talk is based on results proved with David Hemmer (for $d = 2$) and with Bim Gustavsson, Stacey Law, Lorenzo Putignano and Liron Speyer (for $d > 2$).

Yumeng Wu (Peking University)

Title: PBW and Canonical Bases for Tame Quivers: Geometry of Transition Coefficients

Abstract: This talk concerns the geometric meaning of the transition coefficients between PBW bases and the canonical basis of quantum groups associated with tame quivers, with a focus on the Kronecker quiver. After recalling Lusztig’s geometric construction, I will explain a realization of PBW basis elements by sheaf complexes on representation varieties. The coefficients expressing canonical basis elements in the PBW basis can then be described through graded multiplicities of local systems in restrictions of intersection cohomology complexes to suitable strata. I will discuss how purity enters this description and how the resulting geometric interpretation accounts for the triangularity of the transition matrix and the positivity of its coefficients. The talk is based on joint work with Jie Xiao.